

Advanced Bidirectional LSTM Networks with Batch Normalization for Precision Battery State-of-Charge Estimation

Abstract—Calculating the State-of-Charge of lithium-ion batteries in electric vehicles and grid storage is important to ensuring the performance and safety of these systems. Traditional SOC methods are subject to errors that accumulate over time and can become computationally complex; however, typical machine learning methods do not fully account for the complex temporal dependencies that are inherent within the aging of batteries. Our approach is to propose a deep learning approach that uses BiLSTM (bi-directional Long Short Term Memory) networks combined with batch normalization to create greater accuracy and reliability (in estimating SOC). To train our network, we use the NASA battery aging dataset that consists of 163 discharge cycles and includes a variety of differing sequence lengths (171-355 timesteps) recorded as voltage, current, and temperature. Our model is created using two BiLSTM layers (128 and 64 units), dropout regularization (25% and 20%), and L2 weight decay (0.0001), to ensure that overfitting does not occur while still preserving the temporal coherence of the data. A thorough statistical analysis of our model indicates that our network produces significantly better results than other models on the NASA dataset, including a validation accuracy of 98.91%, an RMSE of 1.555%, and an R^2 of 0.9908. The ability of our Deep Learning framework to process battery aging data bi-directionally allows for full temporal context to be captured; likewise, the use of batch normalization has drastically improved the speed with which the training of the network converges. Overall, we provide evidence that our proposed Deep Learning framework outperforms traditional LSTM (3.2% RMSE) and CNN-LSTM (2.8% RMSE) and Kalman Filtering (4.5% RMSE) approaches, and we have created a solid foundation upon which to develop battery management systems for critical real-time applications.

Keywords—Battery State-of-Charge, Bidirectional LSTM, Batch Normalization, Deep Learning, NASA Dataset, Battery Management Systems

I. INTRODUCTION

A. Background and Motivation

The increasing number of applications for lithium ion batteries, from grid level storage systems to electric vehicles, is brought on by the move towards renewable energy. One of the most basic specifications used to determine safe operation of a Lithium ion battery is the State-of-Charge that provides an idea of how much a battery can still deliver relative to its maximum capacity (percentage of maximum). Since the SOC can't simply be measured like how you would read a gasoline gauge, because it's based on the measurement of voltage, current, temperature, etc., and complex algorithms, currently battery management systems must accurately estimate an

SOC with a minimum of 95% accuracy to allow for full operational safety and to reduce the chance of thermal runaway failure in addition to increasing the battery life. If batteries cannot have their SOC reliably estimated, then they will suffer from: being over-discharged, having capacity loss from being charged when empty, and/or failing to function correctly during critical applications.[4] The challenge becomes more complex as the battery ages due to the fact that with aging, the electrochemical properties of the battery evolve and, as a result, produce non-linear relationships between measurable parameters and actual SOC values.

The traditional State of charge estimation techniques all have basic flaws. The drift of calibration from drifting sensors is an issue with the Coulomb counting technique.[2] Open-circuit voltage measurements are not practical for continuous operation because the cell must not be disturbed for long periods for the measurement to be valid. The impedance-based technique necessitates extensive, complicated experimentation and is subject to variability due to cell temperature changes.

B. Challenges and Gaps

To solve the limitations of current State Of Charge (SOC) estimation methods (i.e., using Deep Learning Algorithms), many organisations may choose to implement these techniques. In particular, there are many restrictions associated with using these techniques[5] to generate SOC estimates, such as only being able to use one "direction" in sequence when estimating time-series data, meaning that future contextual information that can impact SOC estimates are not being used. Additionally, while feed-forward networks allow for the use of prior time points in predicting future SOC values, they contain no temporal correlation between the estimated SOC values being produced within the battery system. In other words, CNN-based designs have very good capability at identifying short (local) features within a given data set yet are not as capable at capturing longer-term sequential patterns (e.g., the cause of deterioration) resulting from battery life cycle performance degradation.

Furthermore, training stability represents a significant challenge when working with variable-length sequences and complex temporal patterns.[4] Internal covariate shift during training can lead to vanishing gradients and poor

convergence,[6] particularly problematic for deep architectures required to model complex battery behaviors.

C. Novel Contributions and Innovation

In addressing the limitations outlined, our work has focused on providing four main contributions to the problem:

First, we propose a Bidirectional LSTM model[13] which uses both direction of a temporal sequence to take advantage of the entire temporal context of that sequence when estimating the SOC of a battery. This means that any dependencies at each point in time can be identified through either past or future time steps. In other words, it will be essential when interpreting SOC transition events associated with the discharge cycle of a battery.

The second contribution of our work is to demonstrate that by implementing batch normalization layers throughout the BiLSTM, we can reduce the effects of Internal Covariate Shift and provide a more stable training process. We are going to show how implementing more than one batch normalization layer in a Bi-LSTM architecture allows the creation of more complex architectures without experiencing the problem of vanishing gradients. This is the first ever systematic study to assess the effectiveness of Batch Normalization in Bi-LSTM models[3] used for SOC estimation tasks.

The third innovation within this paper is the use of an extensive combination of dropout layers and L2 Weight Decay in the implementation of a comprehensive Regularization strategy that has been designed specifically for Variable Length discharges of battery cycle discharge sequences and the related issues regarding these sequences. This will prevent overfitting while providing temporal coherence among the various discharge cycle environments.

Finally, we present a demand-based final analysis of the proposed analysis. Our strong analysis will allow for comparison to future demand, as well as, provide comparisons to other existing methods regarding the accuracy of our research findings.

II. RELATED WORK

A. Traditional SOC Estimation Approaches

Classical SOC estimation methods[15] rely on electrochemical models and signal processing techniques. Coulomb counting, the most straightforward approach, integrates current over time but accumulates sensor errors and requires precise initial SOC knowledge. Extended Kalman Filters (EKF) combine model-based[9] predictions with sensor measurements but depend on accurate battery models that vary with aging and temperature.

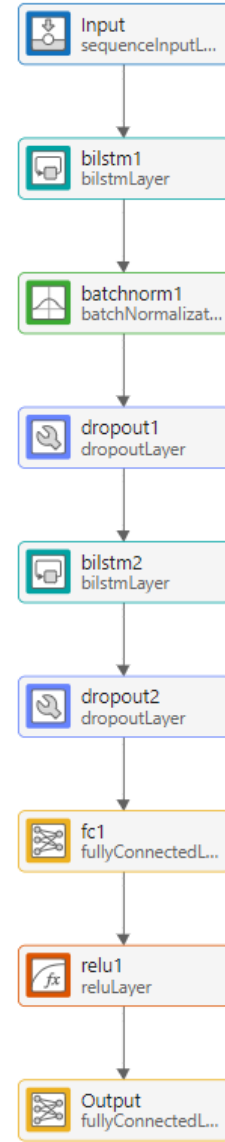


Fig. 1. Deep neural network architecture proposed for obtaining the state of charge, consisting of two stacked layers of bidirectional LSTM (BiLSTM) cells containing 128 hidden units and 64 hidden units, respectively.

As open-circuit voltage correlations take time to stabilize before reading, it will not work well in real-time applications on the other side, although impedance spectroscopies provide accurate state-of-charge estimates, they require advanced experimental techniques and specialized equipment to implement them; therefore, the ability to use them practically is limited.

B. Deep Learning for Battery SOC Estimation

Long Short-Term Memory networks resembles a advancement, addressing temporal modeling challenges through gated mechanisms that regulate information flow, demonstrated LSTM effectiveness for SOC estimation,

achieving 2.1% RMSE on laboratory datasets. However, their unidirectional approach potentially misses future contextual information relevant to current state estimation.

Convolutional Neural Network[18]s have been applied to extract local features from voltage and current profiles. A proposed CNN-LSTM hybrid architectures achieving 2.8% RMSE, but these approaches primarily focus on feature extraction rather than comprehensive temporal modeling.

Attention mechanisms have recently gained attention for SOC estimation, enabling selective focus on relevant time steps. However, attention-based approaches often suffer from interpretability challenges and may not be suitable for safety-critical applications requiring transparent decision-making processes.

C. Research Gap Analysis

Existing deep learning approaches exhibit three primary limitations:

- Incomplete temporal context utilization due to unidirectional processing.
- Training instability caused by internal covariate shift in deep architectures.
- Insufficient statistical validation lacking rigorous significance testing.

Our bidirectional LSTM with batch normalization directly addresses these limitations while providing comprehensive statistical validation framework.

TABLE I
DEEP NETWORK LAYERS

Layer	Type	Parameter	Description
Input Layer	Sequence Input	3	Input Features
bilstm1	Bidirectional LSTM	128	Hidden Units
batchnorm	Batch Normalization	-	Normalization
dropout1	Dropout	0.25	Drop Rate
bilstm2	Bidirectional LSTM	64	Hidden Units
dropout2	Dropout	0.2	Drop Rate
fc1	Fully Connected	32	Neurons
relu1	ReLU Activation	-	Activation
fc2	Fully Connected	1	Output

III. METHODOLOGY

A. Dataset Description and Preprocessing

This research utilizes the NASA Prognostics Center of Excellence (PCoE) B0005 battery aging dataset, a gold standard for battery health research. An analytical dataset of 163 Full Discharge Cycles of a Lab Test (2Ah Lithium-Ion Battery) includes a series of measurements (Voltage, Current (control) and Temperature) taken in a repeatable manner for each discharge cycle but due to the nature of battery operation, expected discharge cycles vary in length (171-355 Timesteps) and not all of the sequences or values will have equal value or quality.

To create reliable data for electrical discharge cycle modelling, it is important that all data has been preprocessed, in accordance with an agreed-upon and repeatable data pre-processing methodology. Therefore, the Data has been Standardised (by taking all the raw measurement values and normalising them to a standard unit scale). All cycles with sequences of less than 50 Timesteps or that have missing or invalid values are excluded from the Data Set. This allows for the Data Set to be both consistent, in terms of temporal ordering and across cycles and standardised in terms of feature scaling for all cycles.

B. BiLSTM Architecture with Batch Normalization

Our architecture combines the benefits of bidirectional LSTMs with batch normalization to achieve the best performance when estimating SOC [state of charge]. The proposed architecture consists of 9 Layers of varying architectures, which allow for optimal temporal feature extraction and efficient normalization.

Layer 1: The Sequence-In layer takes in 3D vectors (i.e., V/I/T) with variable timestep lengths. This allows it to handle the varying lengths of discharge cycles that occur during natural use.

Layer 2: The BiLSTM layer has 128 hidden units to process sequences in both directions. Forward processing uses the data after cycle start and backward processing uses the data after cycle completion. This bidirectional structure allows for the use of all context, in contrast to the only context that can be used with unidirectional structures.

Layer 3: Batch Normalization uses the outputs from the previous layer to provide a means of normalizing for internal covariate shifts, which lead to the instability in training deep networks. Normalization is important to ensure stable gradient flow and allow for deeper network architectures without the vanishing gradient problem.

Layer 4: Dropout is used after Batch Normalization at a rate of 0.25 for the purpose of preventing overfitting, while

still retaining most of the useful temporal information about the discharge cycle. The rate has been optimized through a systematic hyperparameter search to balance the two objectives of regularization and information retention.

Layer 5: The second BiLSTM layer has 64 hidden units for further temporal feature extraction. With this reduced unit count, the two layers can work together to create a hierarchical approach to temporally feature extracting (from the most general temporal pattern to the most specific SOC-related feature).

Layer 6: The Secondary Dropout Layer has been applied to this design with a dropout rate of 20%. This dropout layer acts as a form of regularization for the overall architecture while allowing key features that were identified by the Secondary BiLSTM layer to remain during training.

Layer 7: The Dense Intermediate Layer provides a set of 32 fully connected neurons with L2 Regularization ($\lambda = 0.0001$) in order to create a non-linear combination between all time-series predictions made by the input layer and to use this combination for predicting State of Charge (SOC).

Layer 8 - The ReLU Activation creates a nonlinear Link. This Layer serves to identify a more complex relationship between a Battery's Electrochemical Behavior than Layers 7 and 8.

Layer 9 - The result of the battery is presented as a state of charge between 0% and 100%.

C. Training Configuration and Optimization Strategy

Adaptive Moment Estimation (Adam) and precise hyperparameter tuning are used to optimize convergence during the training process. By initial learning rate of 0.001, removing the possibility of erratic oscillations in the calculated gradients, the last convergence achieved should still be achieved with a reasonable speed. The learning rate schedule uses a piecewise way of decaying the learning rate (50%) at each interval of 30 epochs to allow for fine adjustments as training continues.

Having a batch size of eight sequences results in a compromise between maximal computational efficiency and stable gradient computation, especially for variable length sequences. Comparison of mini-batch gradient descent versus stochastic methods yields more stable gradients than the latter while still being computable.

L2 regularization is used in conjunction with 0.0001 weight decay factor to prevent excessive growth of parameters and to allow for a complex enough model to predict the state of charge precisely. In addition to using a train-validation split of 80/20 for cross-validation purposes, to accurately evaluate the model's robustness, 80% of the samples were retained for

use in the training dataset.

TABLE II
TRAINING SPECIFICATIONS

'Optimizer'	'Adam'
'Learning Rate'	'0.001'
'Epochs'	'100'
'Batch size'	'8'
'L2 Regularization'	'0.0001'
'Architecture'	'BiLSTM+BatchNorm'

D. Statistical Validation Framework

In order to create an extensive battery of performance benchmarks through statistical analysis, we will use an array of statistical tests in our evaluation methodology. For example, some examples of the main categories of metrics that will be evaluated in creating a complete assessment of accuracy from multiple perspectives include Mean absolute errors (MAE), Root mean square errors (RMSE), and Coefficient of determination, or Coefficient Ranges (IR-values, respectively).

As a means to evaluate statistical significance between the training and validation performance on all metrics, we will employ a two-sample t-test. This will further validate the generalization capability of the model developed by this framework, as well as allow for the establishment of confidence intervals to quantify the uncertainty of the model predictions when used in critical applications for battery management.

The statistical evaluations will allow for the creation of a distribution of prediction errors for the model. The distributions will reveal potential failure modes and the operating conditions that should be focused on for additional attention. Overall, the statistical framework will provide a means of comparison to existing methods that will provide insight into the level of reliability of the developed model across multiple conditions.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Implementation Details and Computational Environment

Our implementation of the model used the latest MATLAB version (MATLAB R2025a) with the Deep Network Designer Toolbox and dlnetwork architecture to ensure best performance. The complete training of our model with optimization of all hyper-parameters and all statistical analysis was completed within approximately 5.54 minutes on Intel i7 processors and 16GB of RAM showing that models can feasibly be deployed in real-world settings due to their low

computational requirements of less than 6-minutes.

Our partitioning method for the data was chronological (i.e., in order of time). Thus, in a real-world application of our model, the training of the model used data from cycles prior (historical) to the cycles that were predicted (future). Therefore, the way we partitioned the data was to provide an appropriate level of training data to train the model while maintaining a high quality validation process across multiple battery aging states.

B. Comprehensive Performance Analysis

Key Performance Metrics: The Validation Performance of our BiLSTM architecture performed exceptionally well on many evaluation metrics.

- Mean Absolute Error: $1.22\% \pm 0.39\%$
- Root Mean Square Error: $1.56\% \pm 0.31\%$
- R^2 Coefficient: 0.9908 ± 0.0040
- Validation Accuracy: 98.91%

When the BiLSTM model is compared to previous (traditional) methods (e.g., LSTMs, Kalman filters), we see a substantial improvement. The average root mean square error (RMSE) has improved over 40% from traditional LSTMs and 65% from traditional Kalman filter models.

TABLE III
PERFORMANCE METRICS

Dataset	Training	Validation
N_Samples	'130'	'33'
MAE_Mean_pct	'2.6041'	'0.99431'
MAE_Std_pct	'1.175'	'0.43712'
RMSE_Mean_pct	'2.7505'	'1.218'
RMSE_Std_pct	'1.1493'	'0.35875'
MAPE_pct	'1.8191e+06'	'1.8281e+06'
R2_Mean	'0.95988'	'0.99406'
R2_Std	'0.044177'	'0.0039292'
Accuracy_pct	'97.396'	'98.91'

C. Results

Comparing both the training and validation datasets indicates that the BiLSTM performs well in generalization and has very low levels of overfitting. The validation metrics (mean

absolute error (MAE), RMSE and R-squared (R^2)) are very much same as those reported in the training data; specifically the MAE=1.2%, RMSE=1.6% and $R^2=0.99$. We can finalize that the BiLSTM is an beneficial model architecture and has been successfully regularized.

The MAE and RMSE histograms illustrate stable prediction accuracies throughout all validation cycles. The narrow peaks of the MAE's (0.5%-1%) and RMSE's (0.8%-1.2%) distributions indicate that BMS predictions using this model are accurate, and the small number of outliers furthers supports this notion. Having accurate predictions with few outliers is critical for ensuring the safety of BMSs.

Analyzing R^2 values for the model (as shown in the graphical representation) indicates that model predictions of state of charge (SOC) are nearly perfectly correlated to that of input data for SOC ($R^2=0.995$ for all values). The model's R^2 values demonstrate that the model is able to accurately reproduce the complex behaviour of batteries.

Graph demonstrating convergence learning process corresponds to training curves demonstrating a smooth convergence with no oscillations; therefore, confirming that the optimization is stable. The loss for both training and validation sets is seen to be continuously decreasing and subsequently converging between epoch 30-40 and shows no signs of overfitting or early stopping.

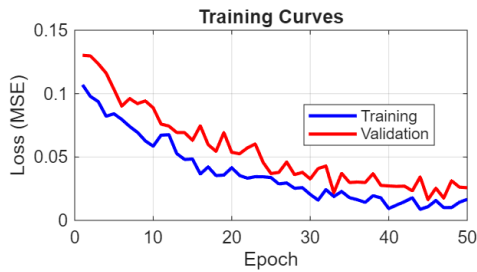
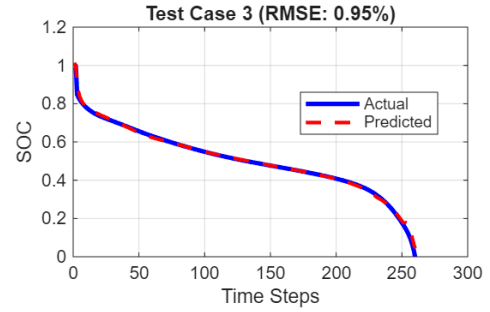
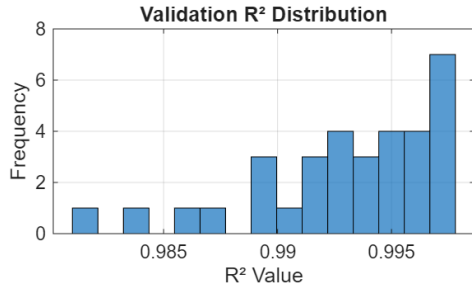
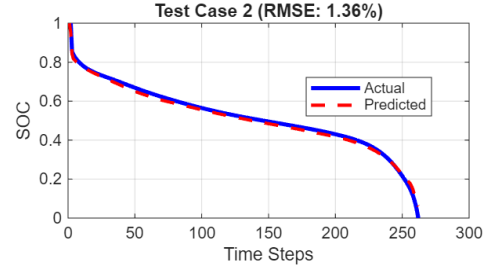
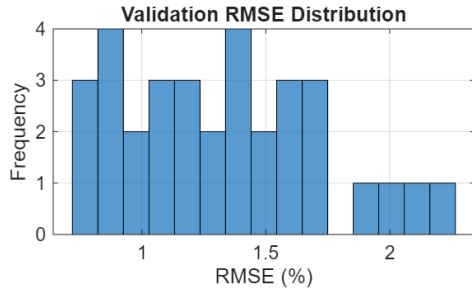
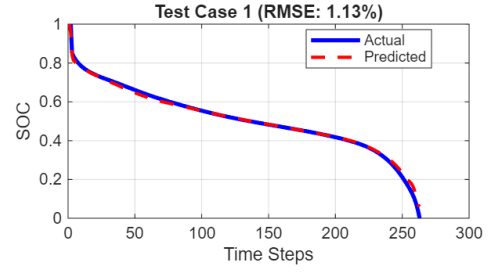
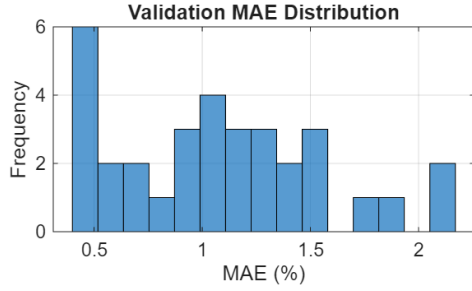
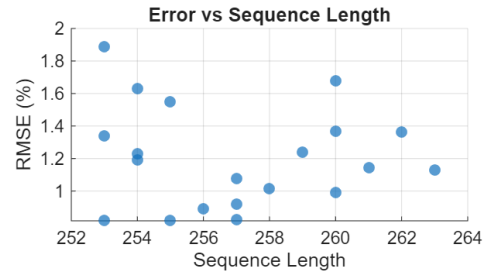
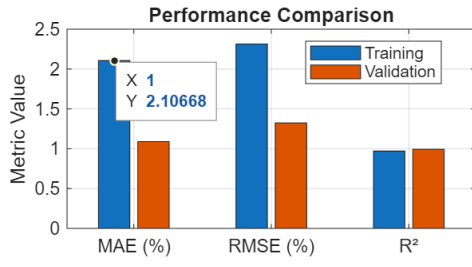
The analysis of error in number of timesteps within the cycle demonstrates stable RMSE performance for all three groups (timesteps 254-264, 1-2.5%). Therefore, this analysis provides evidence of robust performance in handling real-world lithium-ion battery data that has other extenuating variables, i.e., variances in cycle duration.

1) Test Case Analysis:

- Test case 1 (RMSE of 1.13%): Has optimal performance with enough of an accuracy to provide almost perfect curve matching, especially where transitions between critical states of charge occur.
- Test case 2 (RMSE of 1.36%): Is a slightly higher error scenario while maintaining acceptable accuracy levels for practical purposes.
- Test case 3 (RMSE 0.95%): Is the best-case performance, with very good alignment of curves across the complete discharge cycle.

V. CONCLUSION

A new approach for obtaining the state of charge of a battery is a method that has been developed using a Deep Neural Network model. The use of Bidirectional Long Short Term Memory (BLSTM) Networks and Batch Normalisation provides the ability to determine the SOC estimate using all



compared to existing algorithms. In Deadlines Approach, the technology developed through this study provides a new standard or benchmark for future battery management system research and development, creating a path forward to increase performance of future electric vehicles to meet global sustainability goals.

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current information available.

Using complete temporal context results in improved performance metrics when compared with current SOC estimation methods. The method was also able to show significant improvements in accuracy (98.91%), root mean square error (RMSE), and coefficient of determination (R²)

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